

CYBERSTART LEVEL 1

Cybersecurity Foundation Program

Final 16-Lecture Professional Curriculum

A focused foundation program covering cybersecurity, Linux, ethical hacking, network scanning, Fortinet-oriented firewall and network security, cyber defense, practical labs, case studies and career development.

16

lectures

1 HOUR

per lecture

51

course topics

01	Introduction to Cybersecurity
02	Linux Fundamentals
03	Ethical Hacking Basics
04	Firewall & Network Security (Fortinet)
05	Cyber Defense & Threat Prevention
06	Hands-on Practical Labs
07	Real-world Case Studies
08	Career Guidance & Next Steps

Course scope: This is a 16-hour foundation program. Short and overlapping items have been combined where appropriate while retaining distinct practical and security concepts.

16-Lecture Master Schedule

Each lecture is designed for one hour. Practical lectures emphasize supervised lab work, while conceptual lectures focus on core principles, demonstrations and applied discussion.

Lecture	Title	Learning Area	Duration
01	Cybersecurity Fundamentals	Introduction to Cybersecurity	60 min
02	Linux Fundamentals	Linux Fundamentals	60 min
03	Linux Package Management	Linux Fundamentals	60 min
04	Ethical Hacking Foundations	Ethical Hacking Basics	60 min
05	Reconnaissance, Network Scanning & Enumeration	Ethical Hacking Basics	60 min
06	Vulnerability Assessment Practical	Hands-on Practical Labs	60 min
07	Secure Network Architecture	Firewall & Network Security (Fortinet)	60 min
08	Fortinet / FortiGate Firewall Security	Firewall & Network Security (Fortinet)	60 min
09	VPN & Core Network Services Security	Firewall & Network Security (Fortinet)	60 min
10	Malware, Ransomware & Endpoint Defense	Cyber Defense & Threat Prevention	60 min
11	IDS/IPS & Security Monitoring	Cyber Defense & Threat Prevention	60 min
12	Wireshark Network Analysis Practical	Hands-on Practical Labs	60 min
13	Web Application Security Practical	Hands-on Practical Labs	60 min
14	Real-World Cybersecurity Case Studies	Real-world Case Studies	60 min
15	Cybersecurity Career Guidance & Skill Development	Career Guidance & Next Steps	60 min
16	Integrated Practical Assessment	Hands-on Practical Labs / Next Steps	60 min

LECTURE 01 Cybersecurity Fundamentals**Learning Area** Introduction to Cybersecurity**Duration** 60 minutes**1.1** Cybersecurity, Information Security & Ethical Hacking Overview**1.2** CIA Triad - Confidentiality, Integrity & Availability**1.3** Authentication, Authorization, Accountability & Non-Repudiation**1.4** Threats, Vulnerabilities, Risks, Exploits & Attack Surfaces**LECTURE 02 Linux Fundamentals****Learning Area** Linux Fundamentals**Duration** 60 minutes**2.1** Virtualization Overview & Installing Kali Linux on VMware**2.2** Kali Linux Repositories, Users, Updates & Basic System Setup**2.3** Linux Users, Groups, Files, Directories & Permissions**2.4** Linux File System Structure & Essential Directories

LECTURE 03 Linux Package Management**Learning Area** Linux Fundamentals**Duration** 60 minutes

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| 3.1 | Linux Package Management Fundamentals - APT & dpkg |
| 3.2 | Installing, Updating & Upgrading Linux Packages |
| 3.3 | Removing Packages, Dependencies & Package Maintenance |

LECTURE 04 Ethical Hacking Foundations**Learning Area** Ethical Hacking Basics**Duration** 60 minutes

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| 4.1 | Ethical Hacking Ethics, Authorization, Scope & Lab Safety |
| 4.2 | Hacker Types, Attack Categories & Ethical Hacking Phases |
| 4.3 | Black-Box, Gray-Box & White-Box Testing |
| 4.4 | Isolated Ethical Hacking Lab, VM Networking & Snapshots |

LECTURE 05**Reconnaissance, Network Scanning & Enumeration****Learning Area** Ethical Hacking Basics**Duration** 60 minutes

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| 5.1 | Information Gathering, Footprinting & Reconnaissance |
| 5.2 | Network Scanning with Nmap - Host Discovery, Ports, Services & Version Detection |
| 5.3 | Enumeration Fundamentals & Service Identification |
| 5.4 | Nmap NSE & Basic Network Service Enumeration |

LECTURE 06**Vulnerability Assessment Practical****Learning Area** Hands-on Practical Labs**Duration** 60 minutes

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| 6.1 | Vulnerability Scanning vs Vulnerability Assessment vs Penetration Testing |
| 6.2 | Network & Web Vulnerability Discovery with Nmap & Nikto |
| 6.3 | OpenVAS Vulnerability Assessment - Findings Validation, Severity, False Positives, Evidence & Remediation |

LECTURE 07**Secure Network Architecture**

Learning Area	Firewall & Network Security (Fortinet)
Duration	60 minutes

7.1	Secure Network Architecture & Network Segmentation Fundamentals
7.2	VLANs & Access Control Lists - ACLs
7.3	DMZ Architecture & Controlled Service Exposure

LECTURE 08**Fortinet / FortiGate Firewall Security**

Learning Area	Firewall & Network Security (Fortinet)
Duration	60 minutes

8.1	Firewall Types, NGFW Concepts & FortiGate Security Fundamentals
8.2	Firewall Policies, Rule Management & Secure Configuration
8.3	Network Access Control - NAC & Access Security Fundamentals

LECTURE 09**VPN & Core Network Services Security**

Learning Area	Firewall & Network Security (Fortinet)
Duration	60 minutes

9.1	VPN Fundamentals, IPsec/SSL VPN & Secure Remote Access
9.2	DNS Security Fundamentals
9.3	DHCP Security Fundamentals

LECTURE 10**Malware, Ransomware & Endpoint Defense**

Learning Area	Cyber Defense & Threat Prevention
Duration	60 minutes

10.1	Malware & Ransomware Threats - Malware Types & Indicators of Compromise
10.2	Endpoint Protection - Antivirus, EDR/XDR, Application Allowlisting & Security Baselines
10.3	Malware & Ransomware Detection, Prevention & Recovery

LECTURE 11 **IDS/IPS & Security Monitoring****Learning Area** Cyber Defense & Threat Prevention**Duration** 60 minutes**11.1** IDS/IPS Concepts, Traffic Inspection, Snort & Suricata Fundamentals**11.2** Security Logs, Events, Alerts & Monitoring Fundamentals**11.3** Detection Logic, IOC/Behavior-Based Detection & Alert Triage**LECTURE 12** **Wireshark Network Analysis Practical****Learning Area** Hands-on Practical Labs**Duration** 60 minutes**12.1** Wireshark Packet Capture & Network Traffic Analysis Fundamentals**12.2** Capture Filters, Display Filters & Protocol Analysis**12.3** Identifying Suspicious Network Traffic & Documenting Findings

LECTURE 13**Web Application Security Practical**

Learning Area	Hands-on Practical Labs
Duration	60 minutes

13.1	Web Security Lab Setup - DVWA, WebGoat & OWASP Juice Shop
13.2	Web Application Testing with Burp Suite & OWASP ZAP
13.3	Vulnerability Identification, Evidence & Remediation

LECTURE 14**Real-World Cybersecurity Case Studies**

Learning Area	Real-world Case Studies
Duration	60 minutes

14.1	Vulnerability Assessment & Web Security Case Study
14.2	Network, Firewall & Cyber Defense Case Study
14.3	SOC Incident Response, Lessons Learned & Remediation

LECTURE 15**Cybersecurity Career Guidance & Skill Development**

Learning Area	Career Guidance & Next Steps
Duration	60 minutes
15.1	Cybersecurity Career Paths & Role Selection
15.2	Practical Skill Development - Home Labs, CTFs, Portfolio & GitHub
15.3	Interviews, Certifications & Continuous Learning Roadmap

LECTURE 16**Integrated Practical Assessment**

Learning Area	Hands-on Practical Labs / Next Steps
Duration	60 minutes
16.1	Integrated Cybersecurity Practical - Network Discovery, Service Identification, Vulnerability Assessment & Defensive Review
16.2	Security Analysis, Findings & Remediation Report

Course Delivery & Safety Notes

Total duration	16 hours
Structure	16 lectures, 60 minutes each
Total topics	51
Learning focus	Cybersecurity foundations, Linux, ethical hacking, Fortinet/network security, cyber defense
Practical environment	Use isolated virtual labs and intentionally vulnerable training systems.
Authorization	Only test systems you own or have explicit permission to assess.
Assessment	Integrated final practical with concise findings and remediation reporting.